

Complete syllabus checklist

A clear topic-by-topic progress tracker covering all topics in the AA HL course.

	LEARN		REVIEW		CONFIDENT
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1 Number and algebra		47 lessons		
NO.	LESSON			
1.1	Scientific notation			
1.2.1	Arithmetic sequences and series			
1.2.2	Sigma notation			
1.3	Geometric sequences and series			
1.4.1	Compound interest			
1.4.2	Depreciation			
1.4.3	Inflation			
1.5.1	Laws of exponents			
1.5.2	Introduction to logs			
1.5.3	The number e			
1.6	Simple deductive proofs			
1.7.1	More exponents			
1.7.2	Laws of logarithms			
1.7.3	Change of base law			
1.8.1	Sum of an infinite geometric series			
1.8.2	System of linear equations			
1.8.3	Roots of a polynomial			
1.9.1	Binomial expansion			
1.9.2	Binomial expansion (finding a term)			
1.10.1	Counting principles: product and addition principles			
1.10.2	Factorials and arrangements			
1.10.3	Algebra of factorials			
1.10.4	Combinations and permutations			
1.10.5	Keeping objects together or separated			
1.10.6	Binomial theorem extension			
1.11	Partial fractions			
1.12.1	Complex numbers introduction			

NO.	LESSON			
1.12.2	Operations with complex numbers	○	○	○
1.13.1	Modulus-argument form	○	○	○
1.13.2	Euler form	○	○	○
1.13.3	Complex operations and geometric interpretation	○	○	○
1.13.4	Properties of complex conjugates	○	○	○
1.14.1	Complex solutions to polynomial equations	○	○	○
1.14.2	De Moivre's theorem	○	○	○
1.14.3	Roots of complex numbers	○	○	○
1.14.4	Roots of unity	○	○	○
1.15.1	Proof by induction (intro and series)	○	○	○
1.15.2	Proof by induction (differentiation)	○	○	○
1.15.3	Proof by induction (divisibility)	○	○	○
1.15.4	Proof by induction (inequalities)	○	○	○
1.15.5	Proof by induction (De Moivre)	○	○	○
1.15.6	Proof by contradiction (intro)	○	○	○
1.15.7	Proof by contradiction ($\sqrt{2}$)	○	○	○
1.15.8	Proof by contradiction (infinite primes)	○	○	○
1.15.9	Proof by counter example	○	○	○
1.16.1	System of linear equations (HL only)	○	○	○
1.16.2	System of linear equations part 2	○	○	○

NO.	LESSON			
2.1.1	Gradient of a line	○	○	○
2.1.2	Equation of a straight line	○	○	○
2.1.3	Parallel and perpendicular lines	○	○	○
2.2.1	Functions introduction	○	○	○
2.2.2	Evaluating functions	○	○	○
2.2.3	Inverse functions	○	○	○
2.3	Sketching functions	○	○	○
2.4.1	Features of graphs	○	○	○
2.4.2	Asymptotes	○	○	○
2.5.1	Composite functions	○	○	○
2.5.2	Finding inverse functions	○	○	○
2.6.1	Three forms of a quadratic	○	○	○
2.6.2	Factorising quadratics	○	○	○
2.6.3	Vertex form	○	○	○
2.7.1	Solving quadratic equations	○	○	○
2.7.2	Quadratic inequalities	○	○	○
2.7.3	The discriminant	○	○	○
2.8	Rational functions	○	○	○
2.9.1	Exponential modelling	○	○	○
2.9.2	Logarithmic functions	○	○	○
2.10.1	Solving equations using substitution	○	○	○
2.10.2	Solving equations using gcd	○	○	○
2.11.1	Transformations	○	○	○
2.11.2	Composite transformations	○	○	○
2.12.1	Dividing polynomials	○	○	○
2.12.2	Factor and remainder theorems	○	○	○
2.12.3	Graphs of polynomials	○	○	○
2.12.4	Sum and product of roots	○	○	○
2.13.1	Rational functions with quadratics (in denominator)	○	○	○
2.13.2	Rational functions with quadratics (in numerator)	○	○	○
2.14	Even and odd functions	○	○	○
2.15	Solving inequalities	○	○	○
2.16.1	Graphs of modulus functions	○	○	○
2.16.2	Modulus equations and inequalities	○	○	○
2.16.3	Reciprocal transformations	○	○	○
2.16.4	Graphs of $y = [f(x)]^2$	○	○	○

NO.	LESSON			
3.1.1	Midpoint of a line segment	○	○	○
3.1.2	Distance between two points	○	○	○
3.1.3	Volume and surface area	○	○	○
3.1.4	Trigonometry in 3D	○	○	○
3.2.1	Pythagoras' theorem	○	○	○
3.2.2	SOH CAH TOA	○	○	○
3.2.3	Sine rule	○	○	○
3.2.4	Cosine rule	○	○	○
3.2.5	Area of a triangle	○	○	○
3.3.1	Bearings	○	○	○
3.3.2	Angles of elevation and depression	○	○	○
3.4.1	Radians	○	○	○
3.4.2	Arcs and sectors	○	○	○
3.5.1	Mr Flynn's exact triangles	○	○	○
3.5.2	Unit circle	○	○	○
3.5.3	The CAST diagram	○	○	○
3.5.4	Graphs of sin, cos and tan	○	○	○
3.5.5	Sine rule ambiguous case	○	○	○
3.6	Trigonometric identities	○	○	○
3.7.1	Sinusoidal graphs	○	○	○
3.7.2	Sinusoidal modelling	○	○	○
3.8.1	Solving trigonometric equations	○	○	○
3.8.2	Solving trigonometric equations with identities	○	○	○
3.9.1	Reciprocal trigonometric identities	○	○	○

NO.	LESSON			
3.9.2	Inverse trigonometric functions	○	○	○
3.10	Compound angle identities	○	○	○
3.11	Symmetry in trig functions	○	○	○
3.12.1	Vectors introduction	○	○	○
3.12.2	Operations with vectors	○	○	○
3.12.3	Magnitude and unit vectors	○	○	○
3.13.1	Scalar product and angle between vectors	○	○	○
3.13.2	Properties of the scalar product	○	○	○
3.14.1	Vector equation of a line	○	○	○
3.14.2	Parametric form and Cartesian equations of a line	○	○	○
3.14.3	Angle between two lines	○	○	○
3.14.4	Vectors and kinematics	○	○	○
3.15.1	Intersecting lines	○	○	○
3.15.2	Distance between a point and a line	○	○	○
3.16.1	Vector product	○	○	○
3.16.2	Properties of the vector product	○	○	○
3.16.3	Areas using the vector product	○	○	○
3.17.1	Vector equation of a plane	○	○	○
3.17.2	Cartesian equation of a plane	○	○	○
3.18.1	Intersection between a line and a plane	○	○	○
3.18.2	Intersection between two planes	○	○	○
3.18.3	Intersection of three planes	○	○	○
3.18.4	Angle between a line and a plane	○	○	○
3.18.5	Angle between two planes	○	○	○

NO.	LESSON			
4.1.1	Discrete vs continuous data	○	○	○
4.1.2	Sampling techniques	○	○	○
4.2.1	Histograms and cumulative frequency curves	○	○	○
4.2.2	Box and whisker diagrams	○	○	○
4.2.3	Interpreting box plots	○	○	○
4.2.4	Outliers	○	○	○
4.3.1	Averages and spread	○	○	○
4.3.2	Frequency tables	○	○	○
4.3.3	Constant changes in data	○	○	○
4.3.4	Variance and standard deviation by hand	○	○	○
4.4	Linear regression and correlation	○	○	○
4.5	Probability formulae	○	○	○
4.6.1	Venn diagrams	○	○	○
4.6.2	Tree diagrams	○	○	○
4.7	Discrete random variables	○	○	○
4.8	Binomial distribution	○	○	○
4.9	Normal distribution	○	○	○
4.10	Regression line of x on y	○	○	○
4.11	Testing for independence	○	○	○
4.12	Normal distribution (finding mean and standard deviation)	○	○	○
4.13	Bayes' theorem	○	○	○
4.14.1	Variance and standard deviation	○	○	○
4.14.2	Variance and standard deviation of a discrete random variable	○	○	○
4.14.3	Introduction to continuous random variables	○	○	○
4.14.4	Mean of a continuous random variable	○	○	○
4.14.5	Median and mode of a discrete random variable	○	○	○
4.14.6	Median and mode of a continuous random variable	○	○	○
4.14.7	Variance of a continuous random variable	○	○	○
4.14.8	Linear transformations of X	○	○	○

NO.	LESSON			
5.1.1	Introduction to differentiation	○	○	○
5.1.2	Introduction to limits	○	○	○
5.2	Increasing and decreasing functions	○	○	○
5.3.1	Differentiation power rule	○	○	○
5.3.2	Derivative at a point	○	○	○
5.4	Tangents and normals	○	○	○
5.5.1	Integration (power rule)	○	○	○
5.5.2	Finding $f(x)$	○	○	○
5.5.3	Definite integration	○	○	○
5.5.4	Area under the curve	○	○	○
5.5.5	Finding a limit	○	○	○
5.6.1	Differentiating trigonometric, e^x and $\ln x$	○	○	○
5.6.2	Chain rule	○	○	○
5.6.3	Product rule	○	○	○
5.6.4	Quotient rule	○	○	○
5.7.1	Relationship between graph and derivatives	○	○	○
5.7.2	Sketching derivatives	○	○	○
5.7.3	Sketching derivatives with GDC	○	○	○
5.8.1	Classifying stationary points	○	○	○
5.8.2	Finding points of inflexion	○	○	○
5.8.3	Optimisation	○	○	○
5.9	Kinematics	○	○	○
5.10.1	Integrating e^x and $\frac{1}{x}$	○	○	○
5.10.2	Integrating $\sin$ and $\cos$	○	○	○
5.10.3	Inverse chain rule	○	○	○
5.10.4	Integration by substitution	○	○	○
5.11	Area between two curves	○	○	○
5.12.1	Continuous functions	○	○	○
5.12.2	Differentiable functions	○	○	○

NO.	LESSON			
5.12.3	Differentiation from first principles	○	○	○
5.12.4	Higher derivatives	○	○	○
5.13	L'Hopital's rule	○	○	○
5.14.1	Implicit differentiation	○	○	○
5.14.2	Related rates of change	○	○	○
5.15.1	More derivative rules	○	○	○
5.15.2	More integration rules	○	○	○
5.15.3	Integration using partial fractions	○	○	○
5.16.1	Integration by substitution (HL only)	○	○	○
5.16.2	Integration by parts	○	○	○
5.16.3	Repeated integration by parts	○	○	○
5.17.1	Area between a curve and the y-axis	○	○	○
5.17.2	Volume of revolution (introduction and x-axis)	○	○	○
5.17.3	Volume of revolution (y-axis)	○	○	○
5.17.4	Volume of revolution between two curves	○	○	○
5.18.1	Differential equations introduction	○	○	○
5.18.2	Separation of variables	○	○	○
5.18.3	Euler's method introduction	○	○	○
5.18.4	Euler's method example	○	○	○
5.18.5	Homogeneous differential equations	○	○	○
5.18.6	Integrating factor	○	○	○
5.19.1	Maclaurin series introduction	○	○	○
5.19.2	Maclaurin series	○	○	○
5.19.3	Maclaurin series (substitutions and products)	○	○	○
5.19.4	Maclaurin series: binomial series	○	○	○
5.19.5	Maclaurin series (differentiation and integration)	○	○	○
5.19.6	Maclaurin series (sigma notation)	○	○	○
5.19.7	Maclaurin series (differential equations)	○	○	○
5.19.8	Euler's identity	○	○	○